

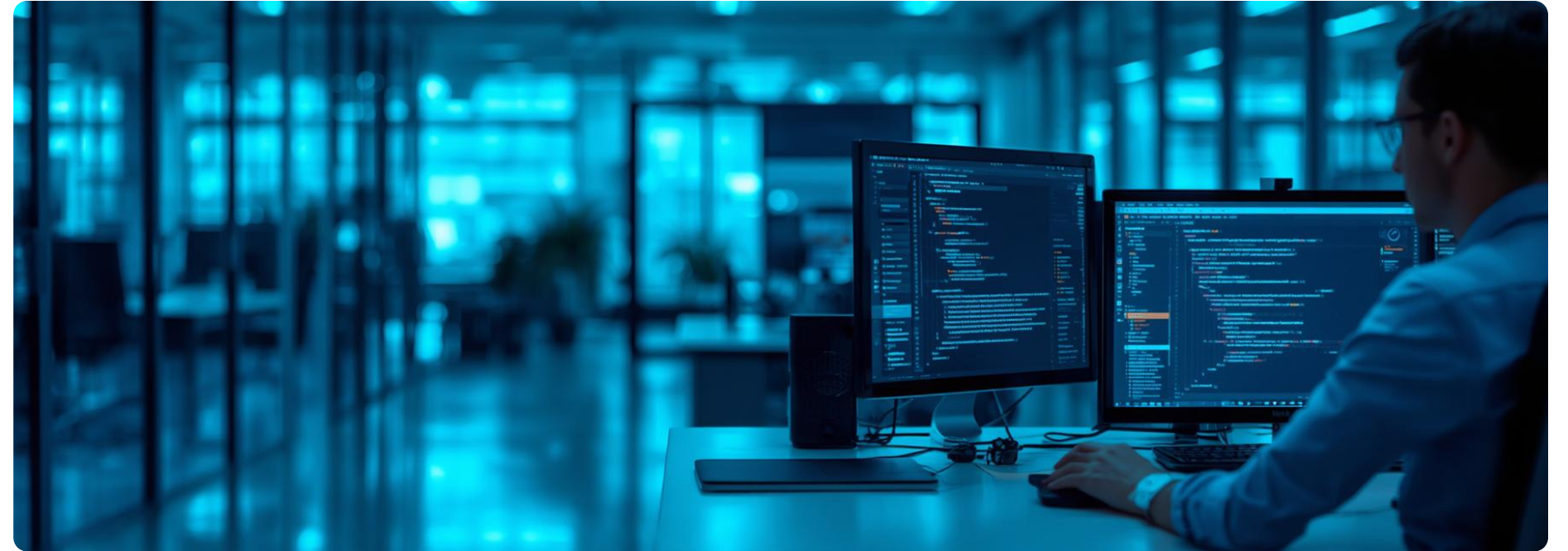


E-commerce Sales Data Analysis

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O7 Solutions IT Company



Company Profile

Founded

2017

Experience

**7+
Years**

Certification

ISO 9001:2015
Certified

Specializations

Mobile, Web,
Software Dev,
Digital Marketing,
IT Integration



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Introduction



E-commerce stores record thousands of transactions daily — this project extracts meaningful business insights.

Tools Used: Python & Pandas for data cleaning; Matplotlib & Seaborn for visualization.

Identifies top products, profitable regions & key customers.

Helps companies make data-driven decisions and increase sales.



Problem Statement

01

Managing large volumes of e-commerce transactional data accurately.

02

Identifying sales trends and customer buying behavior meaningfully.

03

Handling data inconsistencies and integrating diverse data fields.

04

Providing actionable insights quickly for competitive advantage.



Objectives



Trend Analysis

Identify monthly & regional sales trends

Top Products

Identify top categories & products by revenue

Customer Behavior

Analyze buying behavior by region & age group

Order Status

Calculate % of delivered vs returned orders



Technologies Used



Python Libraries

**Pandas &
Seaborn**

Environment

**Jupyter
Notebook**

Data Source

**Kaggle Dataset
1,000 Orders**

Visualization

**Matplotlib,
Seaborn,
Line & Bar Charts**



Methodology

**Idea
Selection**

01

**Data
Collection**

02

**Cleaning &
Preprocessing**

03

**Data
Visualization**

04

**Analysis &
Insights**

05



Screenshots

Order ID	Category	Region	Revenue
1001	Electronics	North	120.50
1002	Electronics	North	150.00
1003	Electronics	North	180.00
1004	Electronics	North	200.00
1005	Electronics	North	220.00
1006	Electronics	North	240.00
1007	Electronics	North	260.00
1008	Electronics	North	280.00
1009	Electronics	North	300.00
1010	Electronics	North	320.00

```
1 // Data cleaning script
2 // Handling missing values and extracting 'Month' column
3
4 // Function to handle missing values
5 function handleMissingValues(data) {
6   // Loop through each row
7   for (let i = 0; i < data.length; i++) {
8     // Check for missing values in 'Revenue' column
9     if (data[i].Revenue === null || data[i].Revenue === '') {
10      // Replace missing values with 0
11      data[i].Revenue = 0;
12    }
13  }
14
15  // Extract 'Month' column for trend analysis
16  const months = data.map(row => row.Month);
17
18  // Return cleaned data and months
19  return { data, months };
20 }
21
22 // Example usage
23 const data = [
24   { Order ID: 1001, Category: 'Electronics', Region: 'North', Revenue: 120.50, Month: 'Jan' },
25   { Order ID: 1002, Category: 'Electronics', Region: 'North', Revenue: 150.00, Month: 'Feb' },
26   { Order ID: 1003, Category: 'Electronics', Region: 'North', Revenue: 180.00, Month: 'Mar' },
27   { Order ID: 1004, Category: 'Electronics', Region: 'North', Revenue: 200.00, Month: 'Apr' },
28   { Order ID: 1005, Category: 'Electronics', Region: 'North', Revenue: 220.00, Month: 'May' },
29   { Order ID: 1006, Category: 'Electronics', Region: 'North', Revenue: 240.00, Month: 'Jun' },
30   { Order ID: 1007, Category: 'Electronics', Region: 'North', Revenue: 260.00, Month: 'Jul' },
31   { Order ID: 1008, Category: 'Electronics', Region: 'North', Revenue: 280.00, Month: 'Aug' },
32   { Order ID: 1009, Category: 'Electronics', Region: 'North', Revenue: 300.00, Month: 'Sep' },
33   { Order ID: 1010, Category: 'Electronics', Region: 'North', Revenue: 320.00, Month: 'Oct' },
34 ];
35
36 // Clean the data
37 const { cleanedData, months } = handleMissingValues(data);
38
39 // Output cleaned data and months
40 console.log('Cleaned Data:', cleanedData);
41 console.log('Months:', months);
```



Dataset preview: sample rows showing order ID, category, region, and revenue fields.

Data cleaning: handling missing values and extracting 'Month' column for trend analysis.

Line chart showing monthly sales trends across the dataset period.

Bar chart (category sales) and pie chart (customer age group distribution).

Conclusion

- » Project successfully analyzed e-commerce sales data to generate actionable business insights.
- » Identified important sales trends and customer behavior patterns across product categories.
- » Leveraged Python, Pandas, and visualization libraries effectively for data processing.
- » Enhanced understanding of data cleaning and visualization techniques for analytical accuracy.





Future Scope

01

Integrate real-time analytics dashboards for live monitoring.

02

Expand to multi-channel sales data: app, web & offline.

03

Apply ML models for sales forecasting & customer segmentation.

04

Customize dashboards for different stakeholder needs.



**Thank
You!**